

Introduction

The Real-Time Public Sentiment Dashboard is a data analytics project designed to monitor and analyze public opinion on social media platforms. By leveraging live Twitter data, the system provides insights into how users perceive a particular brand, product, or topic. This project integrates natural language processing and real-time data streaming to help organizations make informed decisions based on current trends.

Abstract

This project focuses on collecting real-time tweets using the Twitter API and analyzing them using sentiment analysis techniques. The data is processed and classified into positive, negative, and neutral sentiments using the VADER model from NLTK. The results are visualized using a Streamlit dashboard, enabling dynamic and interactive analysis. The system helps in identifying trends, monitoring brand perception, and detecting sudden changes in public sentiment.

Tools Used

Python (Pandas, NLTK, Tweepy), Streamlit for dashboard development, Twitter API for real-time data streaming, and VS Code for development environment.

Steps Involved in Building the Project

The project begins with setting up access to the Twitter API and streaming tweets based on specific keywords. The collected tweets are cleaned by removing unnecessary elements such as links, mentions, and special characters. Sentiment analysis is then applied using the VADER model to classify tweets into different sentiment categories. The processed data is stored and continuously updated. A Streamlit dashboard is developed to visualize the sentiment distribution and display live tweets. The system updates automatically to reflect new incoming data.

Conclusion

The Real-Time Public Sentiment Dashboard successfully demonstrates how social media data can be used to extract meaningful insights in real time. It provides organizations with a powerful tool to monitor public opinion, evaluate marketing strategies, and respond quickly to potential issues. The project highlights the importance of data-driven decision-making and showcases the integration of machine learning, data streaming, and interactive visualization.